Russell Dudek

EY

PHYSICAL AI ENGINEERING

Physical AI Delivery Leader | Digital Twins, Robotics, Machine Vision & High-Reliability Operations

Pittsburgh, Pennsylvania | 412.287.8640 | russelldudek@gmail.com | linkedin.com/in/russelldudek

Candidate vision: https://russelldudek.github.io/EY/

Operator-technologist who turns emerging physical-AI capability into operating systems clients can trust.

Experience spans digital twins, warehouse robotics, machine vision, regulated autonomy, advanced electronics, scaled operations, technical workstreams and AI-enabled decision design.

Manager-level value

Translate engineering evidence into client decisions, operating ownership, delivery cadence and reusable practice assets.

CORE CAPABILITIES

- **Physical AI delivery**
- **Digital twins & simulation**
- **Robotics operations**
- **Machine vision**
- **Technical workstreams**
- **High-reliability engineering**
- **Client-operating translation**
- **AI governance & adoption**
- **Lean / Gemba / Six Sigma**

SELECTED CREDENTIALS

- Google AI Essentials
- IBM Enterprise Design Thinking Practitioner
- Six Sigma Certification
- OSHA 10

TECHNICAL TOOLKIT

Agentic AI | LLM workflows | RAG | Human-in-the-loop design | Python | SQL | JavaScript | TensorFlow | Tableau | Odoo/ERP | Jira | Confluence | HubSpot | GIS

OPERATING METHODS

Decision records, evidence gates, root cause, standard work, scorecards, escalation design, training, adoption, risk controls and continuous learning.

PROFESSIONAL EXPERIENCE

Vape-Jet | Director of Operations — 2025-Present

Leads AI-first operating transformation in a software-enabled robotics and machine-vision environment.

- Connects production, purchasing, quality, support, engineering issue flow, ERP/Odoo, Jira, HubSpot, Slack, telemetry and AI-assisted workflows into one operating system.
- Uses factory, field, customer and machine-fleet signals from approximately 1,000 fielded robots to improve prioritization, risk detection, handoffs, standard work, training and accountability.
- Partners across engineering and operations to turn recurring machine and workflow issues into traceable decisions and sustained operating improvements.

DudeWorth | Founder / AI & Automation Advisor — 2017-Present

Designs practical AI, automation and digital-twin approaches for complex operating environments.

- Develops AI-readiness and opportunity frameworks across value, feasibility, data readiness, governance, adoption burden, reuse and time-to-value.
- Applies digital-twin thinking to warehouse automation, robotic pick/pack/stow, conveyance and material-flow decision support.
- Designs agentic workflows with structured context, retrieval, human judgment, escalation, evaluation and governance rails.

ZeusVu | Chief Pilot, Autonomous Systems & Aerial Data Operations — 2018-2022

- Led regulated drone operations across medical, energy, logistics, real-estate and restricted-airspace contexts with a 100% safety record and no FAA incidents.
- Developed TensorFlow-based concepts for automated aerial inventory and computer-vision-enabled field intelligence.
- Designed medical-drone delivery and chain-of-custody concepts connecting technology, risk, stakeholders and governed execution.

Amazon | Operations Management / Site Leader — 2014-2018

- Led launch and 24/7 execution for Amazon Prime Pittsburgh, supporting approximately five million second-day orders annually.
- Built labor planning, throughput visibility, escalation paths, Gemba routines and standard work across safety, quality and customer experience.
- Contributed to logistics innovation through Prime Air research and Amazon Flex geolocation concepts.

Compunetics | Director of Operations & Engineering Management — 2000-2013

- Progressed through R&D program management, production management, strategic acquisition, operations management and director-level leadership.
- Led engineering, manufacturing, quality, customer delivery and complex technical programs in advanced electronics and high-reliability environments.
- Built quality systems, standard work, root-cause practices, performance measures and process controls.
- Collaborated with high-consequence customers and partners including DoD, CERN, Intel and AMD; published industry articles, delivered technical talks and participated in the HyperTransport Consortium.

DIGITAL-TWIN PRACTICE

Simulation-led decisions for robotic flow

Warehouse automation, robotic pick/pack/stow, conveyance and material-flow decision support - connected to real operating constraints, evidence and ownership.

Virtual-to-physical

Digital twins, simulation logic, machine vision and real-world operating evidence.

Physical-to-operating

Authority, safety, exception handling, maintainability, training and ownership.

Engagement-to-practice

Reusable mechanisms, decision records, delivery economics and team development.

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DIFFERENTIATED TECHNICAL FIT

Digital twins are part of the experience, not a keyword bridge.

Russell has applied digital-twin models and simulation-led thinking to warehouse automation, robotic pick/pack/stow, conveyance and material-flow decisions. The virtual model remains accountable to sensing, actuation, human authority and operating economics.

EXPERIENCE & LEADERSHIP

Cardinal Building Products | Regional Director

2021-2023

- Built the company's first remote regional operation and modernized CRM, commercial, fulfillment, logistics, digital-marketing and online-buying workflows.
- Advanced AI-enhanced transportation-management and route-planning concepts.

Enviro Social Capital | Chairman

Co-founded a public-private initiative focused on sustainable infrastructure and green finance; helped shape a $200 million Social Impact Bond feasibility concept.

IEEE | Pittsburgh Chapter Chair

Led regional programming connecting industry, academia, electronics, manufacturing, STEM outreach and emerging technology.

PHYSICAL-SYSTEMS FOUNDATION

- Software-enabled robotics and machine vision
- Regulated autonomous systems
- Advanced electronics and high-reliability environments
- Warehouse automation, AMRs, ASRS, cobots and transportation systems
- Quality systems, controls and root cause

CLIENT-DELIVERY PROPOSITION

Lead the seam between specialist engineering and client operation: clarify mission, evidence gates, decision rights, risks, run-state ownership and measures required to move from demonstration to sustained value.

PLATFORM ADAPTABILITY & EDUCATION

ROS, Omniverse, MLflow and related environments are implementation platforms to learn and apply within an established physical-systems, digital-twin and delivery discipline.

University of Pittsburgh | B.S.

Materials science, physical chemistry and biology-related studies. Academic work: scholar.google.com/citations?user=yHQSVd8AAAAJ&hl=en.

SELECTED PHYSICAL AI TRANSFER EVIDENCE

SYSTEM CONTEXT	VERIFIED EVIDENCE	RELEVANCE TO EY
Digital twins & material flow	Warehouse automation, robotic pick/pack/stow, conveyance and material-flow decision support.	Links simulation and operational design to real deployment choices.
Fielded robotics	Current operating leadership around approximately 1,000 software-enabled robots and machine-vision workflows.	Brings fleet evidence, issue flow and operating ownership into the workstream.
Regulated autonomy	Autonomous-drone operations with a 100% safety record and no FAA incidents.	Grounds autonomy in authority, fallback, risk and disciplined execution.
High-reliability systems	Advanced electronics, quality systems, root cause and technical programs involving DoD, CERN, Intel and AMD.	Supports rigorous validation, controls and client confidence.
Scaled operations	Amazon Prime launch and 24/7 execution supporting approximately five million second-day orders annually.	Connects technical capability to cadence, throughput, people and customer outcomes.

Close the gap between model capability and client trust.

Connect digital-twin logic, physical evidence, human authority, deployment discipline and operating ownership into a workstream clients can understand, govern and sustain.

VIRTUAL MODEL > PHYSICAL EVIDENCE > CLIENT DECISION > OPERATING OWNERSHIP